A-A-54447
19 July 1991

COMMERCIAL ITEM DESCRIPTION

SEALANT KIT, Dental, Pit and Fissure

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers a dental pit and fissure sealant kit capable of sealing etched pits, grooves, and fissures in teeth to prevent dental caries.

Salient characteristics:

The kit shall be an auto-polymerizing, opaque (off-white) in color, non-toxic pit and fissure sealant kit. The kit shall contain 2.7 milliliters (ml) of universal resin, 2.7 ml of catalyst resin, 7 mixing well trays, 65 syringe insert tips, 7 ml of etchant, 10 disposable mixing sticks, 1 syringe applicator, and instructions for use. The sealant shall contain a bis-GMA resin. This is 2,2-bis-[4-(3-methacryloxy-2-hydroxypropyl)phenyl]-propane.

Universal resin. The universal resin shall contain a minimum of 83% monomers, primarily bis-GMA resin with triethyleneglycol dimethacrylate and bis-phenol-A-dimethacrylate. The remainder of the universal resin shall be composed of the following additives: phenolic stabilizers, aromatic tertiary amine accelerator and pigments.

Catalyst resin. The catalyst resin shall contain a minimum of 97% monomers, primarily bis-GMA resin with triethyleneglycol dimethacrylate and bis-phenol-A-dimethacrylate. The remainder of the catalyst resin shall be composed of the following additives: phenolic stabilizers and diaryl peroxide.

Etchant. The etchant shall be 35.0 weight percent buffered aqueous ortho-phosphoric acid.

Applicator. The applicator shall be made of acrylonitrile-butadiene-styrene (ABS). The applicator shall be 6 ± 1/2 inches in length and thin for ease of handling and shall terminate in a head. The head shall house the piston machinery except for the lever control.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Syringe tips. The syringe insert tips shall be made of polypropylene and shall fit the applicator for the purpose intended. The tips shall be 3/4 inch ± 1/16 inch in length and shall have an inside diameter of 1/16 inch ± 1/64 inch.

Mixing wells. The mixing well trays shall each have 10 mixing wells, shall be made of polypropylene and shall be suitable for the purpose intended. The trays shall be 4 1/4 inches ± 1/4 inch in length and 2 inches ± 1/4 inch in width.

Mixing sticks. The mixing sticks shall be made of acetal copolymer and shall be suitable for the purpose intended. The sticks shall be 4 3/4 inches ± 1/4 inch in length.

Instructions for use. Instructions for use shall be clear and complete and shall describe the mixing of the universal and catalyst resins, operation of the applicator, and the use of the etchant. Mixing the two resins together shall initiate the polymerization reaction. The mixed sealant shall have a working time of 60 to 100 seconds and a maximum setting time of 2.25 minutes when tested as specified below. The sealant shall be easily and conveniently applied by means of a syringe type applicator featuring disposable syringe insert tips.

The sealant shall seal pits, grooves, and fissures in conjunction with the acid etch technique so as to resist dental caries. The sealant shall be retained by the pits and fissures for a minimum of 4 years. The sealant shall be an unfilled resin system which will wear away in a maximum of 48 hours if inadvertently applied to the areas of direct occlusal contact. The sealant kit shall be recognized as acceptable or certified by the Council on Dental Materials, Instruments and Equipment of the American Dental Association.

Certification. The manufacturer shall furnish a statement to the procuring activity, prior to award of contract or purchase order, that the sealant has been tested clinically and found effective in sealing pits, grooves and fissures in teeth to resist dental caries, and that the sealant is free from toxicity as required by this specification. Supporting data shall be forwarded with the statement or made available on request of the procuring activity. If the data furnished is found inadequate, additional information shall be furnished on request. The Government reserves the right to examine the data, to consult with the testing activity, and to determine whether such data is satisfactory and acceptable.

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Immediate containers. Immediate containers for the universal resin, catalyst resin, and etchant shall be plastic squeeze bottles with screw cap closure.

Marking. Unit package (box containing sealant kit) shall be marked with the lot number and date of manufacture and the legend "REFRIGERATE". Immediate containers shall be marked "UNIVERSAL", "CATALYST" or "ETCHING LIQUID" as appropriate.

Delivery. Not more than 3 months shall have elapsed from the date of manufacture of the sealant kit to the date of delivery to the Government.

Workmanship. The components of the kit shall be free from granules and particulates and any defect which detracts from their appearance or impairs their serviceability.

Unit. Each (EA). One sealant kit, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one sealant kit. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests shall be conducted at inspection level S-1, AQL 1.0.

Examination. Sealant kits shall be examined to determine compliance with specification requirements.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Working time. Test shall be performed with an ambient temperature of 23°C ± 1°C. Universal and catalyst resins are placed in separate 3 ml round cylinder Wheaton bottles with 8 mm extended controlled dropper tips. The materials are allowed to come to ambient temperature for at least one hour.

One drop of universal resin is placed in the well of a mixing tray. One drop of catalyst resin is added and a stopwatch is started immediately. The mixture is stirred with a mixing stick at moderate speed for 15 seconds.

The pool of liquid is left undisturbed and is watched. At the moment a ring begins to form in the bottom of the liquid and the pool contracts, the stopwatch is stopped and the time is noted. This is the working time.

Setting time. The test shall be started as in the test for working time, except that 2 drops of universal resin and 2 drops of catalyst resin shall be used.

After mixing, a slender metal rod shall be repeatedly dipped in the liquid and removed at 10-second intervals. The rod shall be wiped clean each time after being removed from the liquid. When liquid no longer adheres to the rod and the mixture has hardened, the stopwatch is stopped. This is the setting time.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-01-216-6260	Sealant Kit, Dental, Pit and Fissure

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2474
Location: PL#2/sealkit

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54447	2. DOCUMENT DATE (YYMMDD) 19 July 1991

3. DOCUMENT TITLE
SEALANT KIT, Dental, Pit and Fissure

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME Defense Personnel Support Center	b. TELEPHONE (Include Area Code) (1) Commercial: (215)737-2870 (2) AUTOVON: 444-2870
c. ADDRESS (Include Zip Code) ATTN: DPSC-MSE 2800 S. 20th Street Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — Previous editions are obsolete

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